

## Planet Fixed Model

5.4

Generated by Doxygen 1.9.1



|                                              |           |
|----------------------------------------------|-----------|
| <b>1 Module Index</b>                        | <b>1</b>  |
| 1.1 Modules                                  | 1         |
| <b>2 Namespace Index</b>                     | <b>3</b>  |
| 2.1 Namespace List                           | 3         |
| <b>3 Hierarchical Index</b>                  | <b>5</b>  |
| 3.1 Class Hierarchy                          | 5         |
| <b>4 Data Structure Index</b>                | <b>7</b>  |
| 4.1 Data Structures                          | 7         |
| <b>5 File Index</b>                          | <b>9</b>  |
| 5.1 File List                                | 9         |
| <b>6 Module Documentation</b>                | <b>11</b> |
| 6.1 Models                                   | 11        |
| 6.1.1 Detailed Description                   | 11        |
| 6.2 Utils                                    | 11        |
| 6.2.1 Detailed Description                   | 11        |
| 6.3 PlanetFixed                              | 11        |
| 6.3.1 Detailed Description                   | 11        |
| 6.4 NorthEastDown                            | 11        |
| 6.4.1 Detailed Description                   | 12        |
| 6.5 PlanetFixedPosn                          | 12        |
| 6.5.1 Detailed Description                   | 12        |
| <b>7 Namespace Documentation</b>             | <b>13</b> |
| 7.1 jeod Namespace Reference                 | 13        |
| 7.1.1 Detailed Description                   | 13        |
| <b>8 Data Structure Documentation</b>        | <b>15</b> |
| 8.1 jeod::AltLatLongState Class Reference    | 15        |
| 8.1.1 Detailed Description                   | 15        |
| 8.1.2 Constructor & Destructor Documentation | 16        |
| 8.1.2.1 AltLatLongState()                    | 16        |
| 8.1.2.2 ~AltLatLongState()                   | 16        |
| 8.1.3 Member Function Documentation          | 16        |
| 8.1.3.1 get_data()                           | 16        |
| 8.1.3.2 JEOD_CLASS_ESTABLISH_FRIENDS2()      | 16        |
| 8.1.3.3 set_data()                           | 17        |
| 8.1.4 Field Documentation                    | 17        |
| 8.1.4.1 altitude                             | 17        |
| 8.1.4.2 latitude                             | 17        |
| 8.1.4.3 longitude                            | 18        |

|                                               |    |
|-----------------------------------------------|----|
| 8.2 jeod::NorthEastDown Class Reference       | 18 |
| 8.2.1 Detailed Description                    | 19 |
| 8.2.2 Member Enumeration Documentation        | 19 |
| 8.2.2.1 AltLatLongType                        | 19 |
| 8.2.3 Constructor & Destructor Documentation  | 19 |
| 8.2.3.1 NorthEastDown() [1/2]                 | 19 |
| 8.2.3.2 ~NorthEastDown()                      | 20 |
| 8.2.3.3 NorthEastDown() [2/2]                 | 20 |
| 8.2.4 Member Function Documentation           | 20 |
| 8.2.4.1 build_ned_orientation()               | 20 |
| 8.2.4.2 JEOD_CLASS_ESTABLISH_FRIENDS2()       | 20 |
| 8.2.4.3 operator=()                           | 20 |
| 8.2.4.4 set_ned_trans_states()                | 20 |
| 8.2.4.5 update_from_cart()                    | 21 |
| 8.2.4.6 update_from_ellip()                   | 21 |
| 8.2.4.7 update_from_spher()                   | 22 |
| 8.2.5 Field Documentation                     | 22 |
| 8.2.5.1 altlatlong_type                       | 22 |
| 8.2.5.2 ned_frame                             | 22 |
| 8.3 jeod::PlanetFixedMessages Class Reference | 23 |
| 8.3.1 Detailed Description                    | 23 |
| 8.3.2 Constructor & Destructor Documentation  | 23 |
| 8.3.2.1 PlanetFixedMessages() [1/2]           | 23 |
| 8.3.2.2 PlanetFixedMessages() [2/2]           | 23 |
| 8.3.3 Member Function Documentation           | 24 |
| 8.3.3.1 JEOD_CLASS_ESTABLISH_FRIENDS2()       | 24 |
| 8.3.3.2 operator=()                           | 24 |
| 8.3.4 Field Documentation                     | 24 |
| 8.3.4.1 domain_error                          | 24 |
| 8.3.4.2 invalid_request                       | 24 |
| 8.4 jeod::PlanetFixedPosition Class Reference | 25 |
| 8.4.1 Detailed Description                    | 26 |
| 8.4.2 Constructor & Destructor Documentation  | 26 |
| 8.4.2.1 PlanetFixedPosition()                 | 26 |
| 8.4.2.2 ~PlanetFixedPosition()                | 26 |
| 8.4.3 Member Function Documentation           | 26 |
| 8.4.3.1 cart_to_ellip()                       | 27 |
| 8.4.3.2 cart_to_spher()                       | 27 |
| 8.4.3.3 ellip_to_cart()                       | 27 |
| 8.4.3.4 get_elliptic_parameters()             | 27 |
| 8.4.3.5 initialize()                          | 27 |
| 8.4.3.6 JEOD_CLASS_ESTABLISH_FRIENDS2()       | 28 |

|                                             |           |
|---------------------------------------------|-----------|
| 8.4.3.7 spher_to_cart()                     | 28        |
| 8.4.3.8 update_from_cart()                  | 28        |
| 8.4.3.9 update_from_ellip()                 | 29        |
| 8.4.3.10 update_from_spher()                | 29        |
| 8.4.4 Field Documentation                   | 29        |
| 8.4.4.1 cart_coords                         | 30        |
| 8.4.4.2 ellip_coords                        | 30        |
| 8.4.4.3 Max_iteration_limit                 | 30        |
| 8.4.4.4 planet                              | 30        |
| 8.4.4.5 Small_radius_limit                  | 31        |
| 8.4.4.6 sphere_coords                       | 31        |
| <b>9 File Documentation</b>                 | <b>33</b> |
| 9.1 alt_lat_long_state.cc File Reference    | 33        |
| 9.1.1 Detailed Description                  | 33        |
| 9.2 alt_lat_long_state.hh File Reference    | 33        |
| 9.2.1 Detailed Description                  | 34        |
| 9.3 class_declarations.hh File Reference    | 34        |
| 9.3.1 Detailed Description                  | 34        |
| 9.4 north_east_down.cc File Reference       | 34        |
| 9.4.1 Detailed Description                  | 34        |
| 9.5 north_east_down.hh File Reference       | 35        |
| 9.5.1 Detailed Description                  | 35        |
| 9.6 planet_fixed_messages.cc File Reference | 35        |
| 9.6.1 Detailed Description                  | 36        |
| 9.6.2 Macro Definition Documentation        | 36        |
| 9.6.2.1 MAKE_PLANETFIXED_MESSAGE_CODE       | 36        |
| 9.7 planet_fixed_messages.hh File Reference | 36        |
| 9.7.1 Detailed Description                  | 36        |
| 9.8 planet_fixed_posn.cc File Reference     | 36        |
| 9.8.1 Detailed Description                  | 37        |
| 9.9 planet_fixed_posn.hh File Reference     | 37        |
| 9.9.1 Detailed Description                  | 37        |
| <b>Index</b>                                | <b>39</b> |



# Chapter 1

## Module Index

### 1.1 Modules

Here is a list of all modules:

|                           |    |
|---------------------------|----|
| Models . . . . .          | 11 |
| Utils . . . . .           | 11 |
| PlanetFixed . . . . .     | 11 |
| NorthEastDown . . . . .   | 11 |
| PlanetFixedPosn . . . . . | 12 |





## Chapter 2

# Namespace Index

### 2.1 Namespace List

Here is a list of all namespaces with brief descriptions:

|                      |                          |                    |
|----------------------|--------------------------|--------------------|
| <a href="#">jeod</a> | Namespace jeod . . . . . | <a href="#">13</a> |
|----------------------|--------------------------|--------------------|



## Chapter 3

# Hierarchical Index

### 3.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

|                                     |    |
|-------------------------------------|----|
| jeod::AltLatLongState . . . . .     | 15 |
| jeod::PlanetFixedMessages . . . . . | 23 |
| jeod::PlanetFixedPosition . . . . . | 25 |
| jeod::NorthEastDown . . . . .       | 18 |



## Chapter 4

# Data Structure Index

### 4.1 Data Structures

Here are the data structures with brief descriptions:

|                                                                                     |    |
|-------------------------------------------------------------------------------------|----|
| <a href="#">jeod::AltLatLongState</a>                                               |    |
| Specifies positional state in planetary altitude, latitude, and longitude . . . . . | 15 |
| <a href="#">jeod::NorthEastDown</a>                                                 |    |
| Defines a local North-East-Down reference frame . . . . .                           | 18 |
| <a href="#">jeod::PlanetFixedMessages</a>                                           |    |
| Specifies the message IDs used in the gravity model . . . . .                       | 23 |
| <a href="#">jeod::PlanetFixedPosition</a>                                           |    |
| Contains various representations of position with respect to a planet . . . . .     | 25 |



## Chapter 5

# File Index

### 5.1 File List

Here is a list of all files with brief descriptions:

|                                                                                                                                         |    |
|-----------------------------------------------------------------------------------------------------------------------------------------|----|
| <a href="#">alt_lat_long_state.cc</a>                                                                                                   |    |
| AltLatLongState class methods . . . . .                                                                                                 | 33 |
| <a href="#">alt_lat_long_state.hh</a>                                                                                                   |    |
| Basic orthogonal Altitude-Latitude-Longitude state definition . . . . .                                                                 | 33 |
| <a href="#">class_declarations.hh</a>                                                                                                   |    |
| Forward declarations of classes defined in planet-fixed position model header files . . . . .                                           | 34 |
| <a href="#">north_east_down.cc</a>                                                                                                      |    |
| NorthEastDown class methods . . . . .                                                                                                   | 34 |
| <a href="#">north_east_down.hh</a>                                                                                                      |    |
| Implementation of the North-East-Down reference frame . . . . .                                                                         | 35 |
| <a href="#">planet_fixed_messages.cc</a>                                                                                                |    |
| Implement the class PlanetFixedMessages . . . . .                                                                                       | 35 |
| <a href="#">planet_fixed_messages.hh</a>                                                                                                |    |
| Define the class PlanetFixedMessages, the class that specifies the message IDs used in the planet-fixed model . . . . .                 | 36 |
| <a href="#">planet_fixed_posn.cc</a>                                                                                                    |    |
| Define PlanetFixedPosition class methods . . . . .                                                                                      | 36 |
| <a href="#">planet_fixed_posn.hh</a>                                                                                                    |    |
| Planet centered fixed position model: alternate coordinate system definitions and transformations to those coordinate systems . . . . . | 37 |





## Chapter 6

# Module Documentation

### 6.1 Models

#### Modules

- [Utils](#)

#### 6.1.1 Detailed Description

### 6.2 Utils

#### Modules

- [PlanetFixed](#)

#### 6.2.1 Detailed Description

### 6.3 PlanetFixed

#### Modules

- [NorthEastDown](#)
- [PlanetFixedPosn](#)

#### 6.3.1 Detailed Description

### 6.4 NorthEastDown

#### Files

- file [north\\_east\\_down.hh](#)  
*Implementation of the North-East-Down reference frame.*
- file [north\\_east\\_down.cc](#)  
*NorthEastDown class methods.*

### 6.4.1 Detailed Description

## 6.5 PlanetFixedPosn

### Files

- file [alt\\_lat\\_long\\_state.hh](#)  
*Basic orthogonal Altitude-Latitude-Longitude state definition.*
- file [class\\_declarations.hh](#)  
*Forward declarations of classes defined in planet-fixed position model header files.*
- file [planet\\_fixed\\_messages.hh](#)  
*Define the class PlanetFixedMessages, the class that specifies the message IDs used in the planet-fixed model.*
- file [planet\\_fixed\\_posn.hh](#)  
*Planet centered fixed position model: alternate coordinate system definitions and transformations to those coordinate systems.*
- file [alt\\_lat\\_long\\_state.cc](#)  
*AltLatLongState class methods.*
- file [planet\\_fixed\\_messages.cc](#)  
*Implement the class PlanetFixedMessages.*
- file [planet\\_fixed\\_posn.cc](#)  
*Define PlanetFixedPosition class methods.*

### 6.5.1 Detailed Description

## Chapter 7

# Namespace Documentation

### 7.1 jeod Namespace Reference

Namespace jeod.

#### Data Structures

- class [NorthEastDown](#)  
*Defines a local North-East-Down reference frame.*
- class [AltLatLongState](#)  
*Specifies positional state in planetary altitude, latitude, and longitude.*
- class [PlanetFixedMessages](#)  
*Specifies the message IDs used in the gravity model.*
- class [PlanetFixedPosition](#)  
*Contains various representations of position with respect to a planet.*

#### 7.1.1 Detailed Description

Namespace jeod.



## Chapter 8

# Data Structure Documentation

### 8.1 jeod::AltLatLongState Class Reference

Specifies positional state in planetary altitude, latitude, and longitude.

```
#include <alt_lat_long_state.hh>
```

#### Public Member Functions

- [AltLatLongState](#) ()=default
- virtual [~AltLatLongState](#) ()=default
- void [set\\_data](#) (double alt, double lat, double lon)  
*Allow user to set member data.*
- void [get\\_data](#) (double &alt, double &lat, double &lon)  
*Allow user to get member data.*

#### Data Fields

- double [altitude](#) {}  
*An object's height above the reference surface of the local GravBody.*
- double [latitude](#) {}  
*An object's north-south angular offset from the local GravBody's reference equator.*
- double [longitude](#) {}  
*An object's east-west angular offset from the local GravBody's reference prime meridian.*

#### Private Member Functions

- [JEOD\\_CLASS\\_ESTABLISH\\_FRIENDS2](#) (jeod, [AltLatLongState](#))

#### 8.1.1 Detailed Description

Specifies positional state in planetary altitude, latitude, and longitude.

Definition at line 77 of file alt\_lat\_long\_state.hh.

## 8.1.2 Constructor & Destructor Documentation

### 8.1.2.1 AltLatLongState()

```
jeod::AltLatLongState::AltLatLongState ( ) [default]
```

### 8.1.2.2 ~AltLatLongState()

```
virtual jeod::AltLatLongState::~~AltLatLongState ( ) [virtual], [default]
```

## 8.1.3 Member Function Documentation

### 8.1.3.1 get\_data()

```
void jeod::AltLatLongState::get_data (
    double & alt_out,
    double & lat_out,
    double & lon_out )
```

Allow user to get member data.

#### Parameters

|     |                |                       |
|-----|----------------|-----------------------|
| out | <i>alt_out</i> | Altitude<br>Units: M  |
| out | <i>lat_out</i> | Longitude<br>Units: r |
| out | <i>lon_out</i> | Latitude<br>Units: r  |

Definition at line 55 of file alt\_lat\_long\_state.cc.

References altitude, latitude, and longitude.

### 8.1.3.2 JEOD\_CLASS\_ESTABLISH\_FRIENDS2()

```
jeod::AltLatLongState::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    AltLatLongState ) [private]
```

### 8.1.3.3 set\_data()

```
void jeod::AltLatLongState::set_data (
    double alt,
    double lat,
    double lon )
```

Allow user to set member data.

#### Parameters

|    |            |                       |
|----|------------|-----------------------|
| in | <i>alt</i> | Altitude<br>Units: M  |
| in | <i>lat</i> | Longitude<br>Units: r |
| in | <i>lon</i> | Latitude<br>Units: r  |

Definition at line 41 of file alt\_lat\_long\_state.cc.

References altitude, latitude, and longitude.

Referenced by jeod::PlanetFixedPosition::update\_from\_ellip(), and jeod::PlanetFixedPosition::update\_from\_spher().

## 8.1.4 Field Documentation

### 8.1.4.1 altitude

```
double jeod::AltLatLongState::altitude {}
```

An object's height above the reference surface of the local GravBody.

trick\_units(m)

Definition at line 84 of file alt\_lat\_long\_state.hh.

Referenced by jeod::PlanetFixedPosition::cart\_to\_ellip(), jeod::PlanetFixedPosition::cart\_to\_spher(), jeod::PlanetFixedPosition::ellip\_to\_cart(), get\_data(), set\_data(), jeod::PlanetFixedPosition::spher\_to\_cart(), jeod::PlanetFixedPosition::update\_from\_ellip(), and jeod::PlanetFixedPosition::update\_from\_spher().

### 8.1.4.2 latitude

```
double jeod::AltLatLongState::latitude {}
```

An object's north-south angular offset from the local GravBody's reference equator.

trick\_units(rad)

Definition at line 90 of file alt\_lat\_long\_state.hh.

Referenced by jeod::NorthEastDown::build\_ned\_orientation(), jeod::PlanetFixedPosition::cart\_to\_ellip(), jeod::PlanetFixedPosition::cart\_to\_spher(), jeod::PlanetFixedPosition::ellip\_to\_cart(), get\_data(), set\_data(), jeod::PlanetFixedPosition::spher\_to\_cart(), jeod::PlanetFixedPosition::update\_from\_ellip(), and jeod::PlanetFixedPosition::update\_from\_spher().

### 8.1.4.3 longitude

```
double jeod::AltLatLongState::longitude {}
```

An object's east-west angular offset from the local GravBody's reference prime meridian.

trick\_units(rad)

Definition at line 96 of file alt\_lat\_long\_state.hh.

Referenced by jeod::NorthEastDown::build\_ned\_orientation(), jeod::PlanetFixedPosition::cart\_to\_ellip(), jeod::PlanetFixedPosition::cart\_to\_spher(), jeod::PlanetFixedPosition::ellip\_to\_cart(), get\_data(), set\_data(), jeod::PlanetFixedPosition::spher\_to\_cart(), jeod::PlanetFixedPosition::update\_from\_ellip(), and jeod::PlanetFixedPosition::update\_from\_spher().

The documentation for this class was generated from the following files:

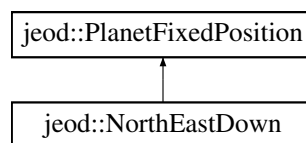
- [alt\\_lat\\_long\\_state.hh](#)
- [alt\\_lat\\_long\\_state.cc](#)

## 8.2 jeod::NorthEastDown Class Reference

Defines a local North-East-Down reference frame.

```
#include <north_east_down.hh>
```

Inheritance diagram for jeod::NorthEastDown:



### Public Types

- enum [AltLatLongType](#) { [undefined](#) = -1 , [spherical](#) , [elliptical](#) }

*Specifies whether the latitude is spherical or elliptical.*

### Public Member Functions

- [NorthEastDown](#) ()=default
- [~NorthEastDown](#) () override=default
- [NorthEastDown](#) (const [NorthEastDown](#) &)=delete
- [NorthEastDown](#) & operator= (const [NorthEastDown](#) &)=delete
- void [update\\_from\\_cart](#) (const double cart[3]) override  
*Update from Cartesian position input.*
- void [update\\_from\\_spher](#) (const [AltLatLongState](#) &spher) override  
*Update from Spherical position input.*
- void [update\\_from\\_ellip](#) (const [AltLatLongState](#) &ellip) override  
*Update from Elliptical position input.*
- virtual void [build\\_ned\\_orientation](#) ()  
*Build NED frame state based on current reference point information.*
- virtual void [set\\_ned\\_trans\\_states](#) (const double pos[3], const double vel[3])  
*Build NED frame state based on current reference point information.*



## Data Fields

- RefFrame [ned\\_frame](#)  
*The local North-East-Down reference frame, centered at the reference point stored in the `ellip_coords`, `sphere_↔coords`, and `cart_coords` data fields inherited from [PlanetFixedPosition](#).*
- [AltLatLongType](#) `altlatlong_type` {undefined}  
*Is reference point specified in spherical or elliptical coords?*

## Private Member Functions

- [JEOD\\_CLASS\\_ESTABLISH\\_FRIENDS2](#) (jeod, [NorthEastDown](#))

## Additional Inherited Members

### 8.2.1 Detailed Description

Defines a local North-East-Down reference frame.

Definition at line 81 of file `north_east_down.hh`.

### 8.2.2 Member Enumeration Documentation

#### 8.2.2.1 AltLatLongType

```
enum jeod::NorthEastDown::AltLatLongType
```

Specifies whether the latitude is spherical or elliptical.

Enumerator

|            |  |
|------------|--|
| undefined  |  |
| spherical  |  |
| elliptical |  |

Definition at line 95 of file `north_east_down.hh`.

### 8.2.3 Constructor & Destructor Documentation

#### 8.2.3.1 NorthEastDown() [1/2]

```
jeod::NorthEastDown::NorthEastDown ( ) [default]
```

### 8.2.3.2 ~NorthEastDown()

```
jeod::NorthEastDown::~~NorthEastDown ( ) [override], [default]
```

### 8.2.3.3 NorthEastDown() [2/2]

```
jeod::NorthEastDown::~NorthEastDown (
    const NorthEastDown & ) [delete]
```

## 8.2.4 Member Function Documentation

### 8.2.4.1 build\_ned\_orientation()

```
void jeod::NorthEastDown::build_ned_orientation ( ) [virtual]
```

Build NED frame state based on current reference point information.

Definition at line 83 of file north\_east\_down.cc.

References [altlatlong\\_type](#), [jeod::PlanetFixedPosition::ellip\\_coords](#), [elliptical](#), [jeod::PlanetFixedMessages::invalid](#), [\\_request](#), [jeod::AltLatLongState::latitude](#), [jeod::AltLatLongState::longitude](#), [ned\\_frame](#), [jeod::PlanetFixedPosition::sphere\\_coords](#), and [spherical](#).

### 8.2.4.2 JEOD\_CLASS\_ESTABLISH\_FRIENDS2()

```
jeod::NorthEastDown::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    NorthEastDown ) [private]
```

### 8.2.4.3 operator=()

```
NorthEastDown& jeod::NorthEastDown::operator= (
    const NorthEastDown & ) [delete]
```

### 8.2.4.4 set\_ned\_trans\_states()

```
void jeod::NorthEastDown::set_ned_trans_states (
    const double pos[3],
    const double vel[3] ) [virtual]
```

Build NED frame state based on current reference point information.

## Parameters

|    |            |                                        |
|----|------------|----------------------------------------|
| in | <i>pos</i> | Cartesian position, PCPF<br>Units: M   |
| in | <i>vel</i> | Cartesian velocity, PCPF<br>Units: M/s |

Definition at line 136 of file north\_east\_down.cc.

References `ned_frame`, and `jeod::PlanetFixedPosition::update_from_cart()`.

## 8.2.4.5 update\_from\_cart()

```
void jeod::NorthEastDown::update_from_cart (
    const double cart[3] ) [override], [virtual]
```

Update from Cartesian position input.

## Parameters

|    |             |                                    |
|----|-------------|------------------------------------|
| in | <i>cart</i> | Cartesian coords, PCPF<br>Units: M |
|----|-------------|------------------------------------|

Reimplemented from [jeod::PlanetFixedPosition](#).

Definition at line 51 of file north\_east\_down.cc.

References `jeod::PlanetFixedPosition::cart_coords`, `ned_frame`, and `jeod::PlanetFixedPosition::update_from_cart()`.

## 8.2.4.6 update\_from\_ellip()

```
void jeod::NorthEastDown::update_from_ellip (
    const AltLatLongState & ellip ) [override], [virtual]
```

Update from Elliptical position input.

## Parameters

|    |              |                                |
|----|--------------|--------------------------------|
| in | <i>ellip</i> | Elliptical AltLatLong position |
|----|--------------|--------------------------------|

Reimplemented from [jeod::PlanetFixedPosition](#).

Definition at line 73 of file north\_east\_down.cc.

References `jeod::PlanetFixedPosition::cart_coords`, `ned_frame`, and `jeod::PlanetFixedPosition::update_from_ellip()`.

#### 8.2.4.7 update\_from\_spher()

```
void jeod::NorthEastDown::update_from_spher (
    const AltLatLongState & spher ) [override], [virtual]
```

Update from Spherical position input.

##### Parameters

|    |              |                               |
|----|--------------|-------------------------------|
| in | <i>spher</i> | Spherical AltLatLong position |
|----|--------------|-------------------------------|

Reimplemented from [jeod::PlanetFixedPosition](#).

Definition at line 62 of file north\_east\_down.cc.

References [jeod::PlanetFixedPosition::cart\\_coords](#), [ned\\_frame](#), and [jeod::PlanetFixedPosition::update\\_from\\_spher\(\)](#).

### 8.2.5 Field Documentation

#### 8.2.5.1 altlatlong\_type

```
AltLatLongType jeod::NorthEastDown::altlatlong_type {undefined}
```

Is reference point specified in spherical or elliptical coords?

trick\_units(-)

Definition at line 112 of file north\_east\_down.hh.

Referenced by [build\\_ned\\_orientation\(\)](#).

#### 8.2.5.2 ned\_frame

```
RefFrame jeod::NorthEastDown::ned_frame
```

The local North-East-Down reference frame, centered at the reference point stored in the [ellip\\_coords](#), [sphere\\_coords](#), and [cart\\_coords](#) data fields inherited from [PlanetFixedPosition](#).

trick\_units(-)

Definition at line 90 of file north\_east\_down.hh.

Referenced by [build\\_ned\\_orientation\(\)](#), [set\\_ned\\_trans\\_states\(\)](#), [update\\_from\\_cart\(\)](#), [update\\_from\\_ellip\(\)](#), and [update\\_from\\_spher\(\)](#).

The documentation for this class was generated from the following files:

- [north\\_east\\_down.hh](#)
- [north\\_east\\_down.cc](#)

## 8.3 jeod::PlanetFixedMessages Class Reference

Specifies the message IDs used in the gravity model.

```
#include <planet_fixed_messages.hh>
```

### Public Member Functions

- [PlanetFixedMessages](#) ()=delete
- [PlanetFixedMessages](#) (const [PlanetFixedMessages](#) &)=delete
- [PlanetFixedMessages](#) & operator= (const [PlanetFixedMessages](#) &)=delete

### Static Public Attributes

- static const char \* [invalid\\_request](#) = "environment/planet\_fixed/" "invalid\_request"  
*Issued when a selection such as an enum value is invalid.*
- static const char \* [domain\\_error](#) = "environment/planet\_fixed/" "domain\_error"  
*Issued when a value is invalid such as an overly small radius.*

### Private Member Functions

- [JEOD\\_CLASS\\_ESTABLISH\\_FRIENDS2](#) (jeod, [PlanetFixedMessages](#))

#### 8.3.1 Detailed Description

Specifies the message IDs used in the gravity model.

Definition at line 83 of file planet\_fixed\_messages.hh.

#### 8.3.2 Constructor & Destructor Documentation

##### 8.3.2.1 PlanetFixedMessages() [1/2]

```
jeod::PlanetFixedMessages::PlanetFixedMessages ( ) [delete]
```

##### 8.3.2.2 PlanetFixedMessages() [2/2]

```
jeod::PlanetFixedMessages::PlanetFixedMessages (
    const PlanetFixedMessages & ) [delete]
```

### 8.3.3 Member Function Documentation

#### 8.3.3.1 JEOD\_CLASS\_ESTABLISH\_FRIENDS2()

```
jeod::PlanetFixedMessages::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    PlanetFixedMessages ) [private]
```

#### 8.3.3.2 operator=()

```
PlanetFixedMessages& jeod::PlanetFixedMessages::operator= (
    const PlanetFixedMessages & ) [delete]
```

### 8.3.4 Field Documentation

#### 8.3.4.1 domain\_error

```
char const * jeod::PlanetFixedMessages::domain_error = "environment/planet_fixed/" "domain_↵
error" [static]
```

Issued when a value is invalid such as an overly small radius.

trick\_units(−)

Definition at line 95 of file planet\_fixed\_messages.hh.

Referenced by jeod::PlanetFixedPosition::cart\_to\_ellip(), and jeod::PlanetFixedPosition::cart\_to\_spher().

#### 8.3.4.2 invalid\_request

```
char const * jeod::PlanetFixedMessages::invalid_request = "environment/planet_fixed/" "invalid_↵
_request" [static]
```

Issued when a selection such as an enum value is invalid.

trick\_units(−)

Definition at line 90 of file planet\_fixed\_messages.hh.

Referenced by jeod::NorthEastDown::build\_ned\_orientation().

The documentation for this class was generated from the following files:

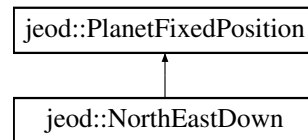
- [planet\\_fixed\\_messages.hh](#)
- [planet\\_fixed\\_messages.cc](#)

## 8.4 jeod::PlanetFixedPosition Class Reference

Contains various representations of position with respect to a planet.

```
#include <planet_fixed_posn.hh>
```

Inheritance diagram for jeod::PlanetFixedPosition:



### Public Member Functions

- [PlanetFixedPosition](#) ()=default
- virtual [~PlanetFixedPosition](#) ()=default
- virtual void [initialize](#) (Planet \*planet\_in)  
*Initialize a [PlanetFixedPosition](#) object.*
- virtual void [update\\_from\\_cart](#) (const double cart[3])  
*Update from Cartesian position input.*
- virtual void [update\\_from\\_spher](#) (const [AltLatLongState](#) &spher)  
*Update from Spherical position input.*
- virtual void [update\\_from\\_ellip](#) (const [AltLatLongState](#) &ellip)  
*Update from Elliptical position input.*

### Data Fields

- [AltLatLongState](#) ellip\_coords  
*An object's current position in elliptical coordinates.*
- [AltLatLongState](#) sphere\_coords  
*The same object's current position in spherical coordinates.*
- double [cart\\_coords](#) [3] {}  
*The planet-centered, planet-fixed position of the object.*
- Planet \* [planet](#) {}  
*The planet currently associated with this.*

### Static Public Attributes

- static constexpr double [Small\\_radius\\_limit](#) {1e-60}  
*Limit of ratio of radial distance to planet equatorial radius below which planetary coordinates are deemed to be invalid.*
- static constexpr int [Max\\_iteration\\_limit](#) {10}  
*Limit of number of iterations used to solve elliptic parameters.*

## Protected Member Functions

- void [cart\\_to\\_spher](#) ()  
*Convert from cartesian to spherical position.*
- void [cart\\_to\\_ellip](#) ()  
*Convert from cartesian to elliptical position.*
- void [spher\\_to\\_cart](#) ()  
*Convert from spherical to cartesian position.*
- void [ellip\\_to\\_cart](#) ()  
*Convert from elliptical to cartesian position.*
- int [get\\_elliptic\\_parameters](#) (double r, double z, double &f, double &h, int maxIters=[Max\\_iteration\\_limit](#))

## Private Member Functions

- [JEOD\\_CLASS\\_ESTABLISH\\_FRIENDS2](#) (jeod, [PlanetFixedPosition](#))

### 8.4.1 Detailed Description

Contains various representations of position with respect to a planet.

Definition at line 91 of file planet\_fixed\_posn.hh.

### 8.4.2 Constructor & Destructor Documentation

#### 8.4.2.1 PlanetFixedPosition()

```
jeod::PlanetFixedPosition::PlanetFixedPosition ( ) [default]
```

#### 8.4.2.2 ~PlanetFixedPosition()

```
virtual jeod::PlanetFixedPosition::~~PlanetFixedPosition ( ) [virtual], [default]
```

### 8.4.3 Member Function Documentation



#### 8.4.3.1 cart\_to\_ellip()

```
void jeod::PlanetFixedPosition::cart_to_ellip ( ) [protected]
```

Convert from cartesian to elliptical position.

Definition at line 136 of file planet\_fixed\_posn.cc.

References `jeod::AltLatLongState::altitude`, `cart_coords`, `jeod::PlanetFixedMessages::domain_error`, `ellip_coords`, `get_elliptic_parameters()`, `jeod::AltLatLongState::latitude`, `jeod::AltLatLongState::longitude`, and `Small_radius_limit`.

Referenced by `update_from_cart()`, and `update_from_spher()`.

#### 8.4.3.2 cart\_to\_spher()

```
void jeod::PlanetFixedPosition::cart_to_spher ( ) [protected]
```

Convert from cartesian to spherical position.

Definition at line 94 of file planet\_fixed\_posn.cc.

References `jeod::AltLatLongState::altitude`, `cart_coords`, `jeod::PlanetFixedMessages::domain_error`, `jeod::AltLatLongState::latitude`, `jeod::AltLatLongState::longitude`, `planet`, `Small_radius_limit`, and `sphere_coords`.

Referenced by `update_from_cart()`, and `update_from_ellip()`.

#### 8.4.3.3 ellip\_to\_cart()

```
void jeod::PlanetFixedPosition::ellip_to_cart ( ) [protected]
```

Convert from elliptical to cartesian position.

Definition at line 208 of file planet\_fixed\_posn.cc.

References `jeod::AltLatLongState::altitude`, `cart_coords`, `ellip_coords`, `jeod::AltLatLongState::latitude`, `jeod::AltLatLongState::longitude`, and `planet`.

Referenced by `update_from_ellip()`.

#### 8.4.3.4 get\_elliptic\_parameters()

```
int jeod::PlanetFixedPosition::get_elliptic_parameters (
    double r,
    double z,
    double & f,
    double & h,
    int maxIters = Max_iteration_limit ) [protected]
```

Definition at line 238 of file planet\_fixed\_posn.cc.

References `planet`.

Referenced by `cart_to_ellip()`.

#### 8.4.3.5 initialize()

```
void jeod::PlanetFixedPosition::initialize (
    Planet * planet_in ) [virtual]
```

Initialize a [PlanetFixedPosition](#) object.

**Parameters**

|    |                               |                   |
|----|-------------------------------|-------------------|
| in | <i>planet</i> ↔<br><i>_in</i> | Associated planet |
|----|-------------------------------|-------------------|

Definition at line 50 of file planet\_fixed\_posn.cc.

References [planet](#).

**8.4.3.6 JEOD\_CLASS\_ESTABLISH\_FRIENDS2()**

```
jeod::PlanetFixedPosition::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    PlanetFixedPosition ) [private]
```

**8.4.3.7 spher\_to\_cart()**

```
void jeod::PlanetFixedPosition::spher_to_cart ( ) [protected]
```

Convert from spherical to cartesian position.

Definition at line 188 of file planet\_fixed\_posn.cc.

References [jeod::AltLatLongState::altitude](#), [cart\\_coords](#), [jeod::AltLatLongState::latitude](#), [jeod::AltLatLongState](#)↔  
[::longitude](#), [planet](#), and [sphere\\_coords](#).

Referenced by [update\\_from\\_spher\(\)](#).

**8.4.3.8 update\_from\_cart()**

```
void jeod::PlanetFixedPosition::update_from_cart (
    const double cart[3] ) [virtual]
```

Update from Cartesian position input.

**Parameters**

|    |             |                                    |
|----|-------------|------------------------------------|
| in | <i>cart</i> | Cartesian coords, PCPF<br>Units: M |
|----|-------------|------------------------------------|

Reimplemented in [jeod::NorthEastDown](#).

Definition at line 59 of file planet\_fixed\_posn.cc.

References `cart_coords`, `cart_to_ellip()`, and `cart_to_spher()`.

Referenced by `jeod::NorthEastDown::set_ned_trans_states()`, and `jeod::NorthEastDown::update_from_cart()`.

#### 8.4.3.9 update\_from\_ellip()

```
void jeod::PlanetFixedPosition::update_from_ellip (
    const AltLatLongState & ellip ) [virtual]
```

Update from Elliptical position input.

##### Parameters

|                 |                    |                                |
|-----------------|--------------------|--------------------------------|
| <code>in</code> | <code>ellip</code> | Elliptical AltLatLong position |
|-----------------|--------------------|--------------------------------|

Reimplemented in [jeod::NorthEastDown](#).

Definition at line 83 of file `planet_fixed_posn.cc`.

References `jeod::AltLatLongState::altitude`, `cart_to_spher()`, `ellip_coords`, `ellip_to_cart()`, `jeod::AltLatLongState::latitude`, `jeod::AltLatLongState::longitude`, and `jeod::AltLatLongState::set_data()`.

Referenced by `jeod::NorthEastDown::update_from_ellip()`.

#### 8.4.3.10 update\_from\_spher()

```
void jeod::PlanetFixedPosition::update_from_spher (
    const AltLatLongState & spher ) [virtual]
```

Update from Spherical position input.

##### Parameters

|                 |                    |                               |
|-----------------|--------------------|-------------------------------|
| <code>in</code> | <code>spher</code> | Spherical AltLatLong position |
|-----------------|--------------------|-------------------------------|

Reimplemented in [jeod::NorthEastDown](#).

Definition at line 71 of file `planet_fixed_posn.cc`.

References `jeod::AltLatLongState::altitude`, `cart_to_ellip()`, `jeod::AltLatLongState::latitude`, `jeod::AltLatLongState::longitude`, `jeod::AltLatLongState::set_data()`, `spher_to_cart()`, and `sphere_coords`.

Referenced by `jeod::NorthEastDown::update_from_spher()`.

### 8.4.4 Field Documentation

#### 8.4.4.1 cart\_coords

```
double jeod::PlanetFixedPosition::cart_coords[3] {}
```

The planet-centered, planet-fixed position of the object.

trick\_units(m)

Definition at line 130 of file planet\_fixed\_posn.hh.

Referenced by cart\_to\_ellip(), cart\_to\_spher(), ellip\_to\_cart(), spher\_to\_cart(), update\_from\_cart(), jeod::NorthEastDown::update\_from\_cart(), jeod::NorthEastDown::update\_from\_ellip(), and jeod::NorthEastDown::update\_from\_spher().

#### 8.4.4.2 ellip\_coords

```
AltLatLongState jeod::PlanetFixedPosition::ellip_coords
```

An object's current position in elliptical coordinates.

Per Vallado p. 140, elliptical latitude is the angle between the equatorial plane and the surface normal on the ellipsoid at the point of interest. Similarly, elliptical longitude is assumed to be the angle between the reference meridian and the surface normal on the ellipsoid at the point of interest. trick\_units(—)

Definition at line 116 of file planet\_fixed\_posn.hh.

Referenced by jeod::NorthEastDown::build\_ned\_orientation(), cart\_to\_ellip(), ellip\_to\_cart(), and update\_from\_ellip().

#### 8.4.4.3 Max\_iteration\_limit

```
constexpr int jeod::PlanetFixedPosition::Max_iteration_limit {10} [static], [constexpr]
```

Limit of number of iterations used to solve elliptic parameters.

trick\_io(\*o) trick\_units(—)

Definition at line 104 of file planet\_fixed\_posn.hh.

#### 8.4.4.4 planet

```
Planet* jeod::PlanetFixedPosition::planet {}
```

The planet currently associated with this.

trick\_units(—)

Definition at line 135 of file planet\_fixed\_posn.hh.

Referenced by cart\_to\_spher(), ellip\_to\_cart(), get\_elliptic\_parameters(), initialize(), and spher\_to\_cart().

#### 8.4.4.5 Small\_radius\_limit

```
constexpr double jeod::PlanetFixedPosition::Small_radius_limit {1e-60} [static], [constexpr]
```

Limit of ratio of radial distance to planet equatorial radius below which planetary coordinates are deemed to be invalid.

trick\_io(\*o) trick\_units(-)

Definition at line 99 of file planet\_fixed\_posn.hh.

Referenced by cart\_to\_ellip(), and cart\_to\_spher().

#### 8.4.4.6 sphere\_coords

```
AltLatLongState jeod::PlanetFixedPosition::sphere_coords
```

The same object's current position in spherical coordinates.

Per Vallado p. 140, spherical latitude is the angle measured at the planet's center from the equatorial plane to the point of interest. Similarly, spherical longitude is also assumed to be the angle measured at the planet's center from the reference meridian to the point of interest. trick\_units(-)

Definition at line 125 of file planet\_fixed\_posn.hh.

Referenced by jeod::NorthEastDown::build\_ned\_orientation(), cart\_to\_spher(), spher\_to\_cart(), and update\_from\_spher().

The documentation for this class was generated from the following files:

- [planet\\_fixed\\_posn.hh](#)
- [planet\\_fixed\\_posn.cc](#)



## Chapter 9

# File Documentation

### 9.1 alt\_lat\_long\_state.cc File Reference

AltLatLongState class methods.

```
#include "../include/alt_lat_long_state.hh"
```

#### Namespaces

- [jeod](#)  
*Namespace jeod.*

#### 9.1.1 Detailed Description

AltLatLongState class methods.

### 9.2 alt\_lat\_long\_state.hh File Reference

Basic orthogonal Altitude-Latitude-Longitude state definition.

```
#include "utils/sim_interface/include/jeod_class.hh"
```

#### Data Structures

- class [jeod::AltLatLongState](#)  
*Specifies positional state in planetary altitude, latitude, and longitude.*

#### Namespaces

- [jeod](#)  
*Namespace jeod.*

### 9.2.1 Detailed Description

Basic orthogonal Altitude-Latitude-Longitude state definition.

## 9.3 class\_declarations.hh File Reference

Forward declarations of classes defined in planet-fixed position model header files.

### Namespaces

- [jeod](#)

*Namespace jeod.*

### 9.3.1 Detailed Description

Forward declarations of classes defined in planet-fixed position model header files.

## 9.4 north\_east\_down.cc File Reference

NorthEastDown class methods.

```
#include <cmath>
#include <cstdio>
#include "utils/math/include/vector3.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/planet_fixed/planet_fixed_posn/include/planet_fixed_messages.↵
hh"
#include "../include/north_east_down.hh"
```

### Namespaces

- [jeod](#)

*Namespace jeod.*

### 9.4.1 Detailed Description

NorthEastDown class methods.



## 9.5 north\_east\_down.hh File Reference

Implementation of the North-East-Down reference frame.

```
#include "utils/ref_frames/include/ref_frame.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "utils/planet_fixed/planet_fixed_posn/include/class_declarations.↵
hh"
#include "utils/planet_fixed/planet_fixed_posn/include/planet_fixed_posn.↵
hh"
```

### Data Structures

- class [jeod::NorthEastDown](#)  
*Defines a local North-East-Down reference frame.*

### Namespaces

- [jeod](#)  
*Namespace jeod.*

#### 9.5.1 Detailed Description

Implementation of the North-East-Down reference frame.

## 9.6 planet\_fixed\_messages.cc File Reference

Implement the class PlanetFixedMessages.

```
#include "utils/message/include/make_message_code.hh"
#include "../include/planet_fixed_messages.hh"
```

### Namespaces

- [jeod](#)  
*Namespace jeod.*

### Macros

- #define [MAKE\\_PLANETFIXED\\_MESSAGE\\_CODE](#)(id) JEOD\_MAKE\_MESSAGE\_CODE(PlanetFixed↵  
Messages, "environment/planet\_fixed/", id)

### 9.6.1 Detailed Description

Implement the class PlanetFixedMessages.

### 9.6.2 Macro Definition Documentation

#### 9.6.2.1 MAKE\_PLANETFIXED\_MESSAGE\_CODE

```
#define MAKE_PLANETFIXED_MESSAGE_CODE (
    id ) JEOD_MAKE_MESSAGE_CODE(PlanetFixedMessages, "environment/planet_fixed/",
    id)
```

Definition at line 44 of file planet\_fixed\_messages.cc.

## 9.7 planet\_fixed\_messages.hh File Reference

Define the class PlanetFixedMessages, the class that specifies the message IDs used in the planet-fixed model.

```
#include "utils/sim_interface/include/jeod_class.hh"
```

### Data Structures

- class [jeod::PlanetFixedMessages](#)  
*Specifies the message IDs used in the gravity model.*

### Namespaces

- [jeod](#)  
*Namespace jeod.*

### 9.7.1 Detailed Description

Define the class PlanetFixedMessages, the class that specifies the message IDs used in the planet-fixed model.

## 9.8 planet\_fixed\_posn.cc File Reference

Define PlanetFixedPosition class methods.

```
#include <cmath>
#include <cstdlib>
#include "environment/planet/include/planet.hh"
#include "utils/math/include/vector3.hh"
#include "utils/message/include/message_handler.hh"
#include "../include/planet_fixed_messages.hh"
#include "../include/planet_fixed_posn.hh"
```

## Namespaces

- [jeod](#)

*Namespace jeod.*

### 9.8.1 Detailed Description

Define PlanetFixedPosition class methods.

## 9.9 planet\_fixed\_posn.hh File Reference

Planet centered fixed position model: alternate coordinate system definitions and transformations to those coordinate systems.

```
#include "environment/planet/include/class_declarations.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "alt_lat_long_state.hh"
#include "environment/planet/include/planet.hh"
```

## Data Structures

- class [jeod::PlanetFixedPosition](#)

*Contains various representations of position with respect to a planet.*

## Namespaces

- [jeod](#)

*Namespace jeod.*

### 9.9.1 Detailed Description

Planet centered fixed position model: alternate coordinate system definitions and transformations to those coordinate systems.



# Index

- ~AltLatLongState
  - jeod::AltLatLongState, [16](#)
- ~NorthEastDown
  - jeod::NorthEastDown, [19](#)
- ~PlanetFixedPosition
  - jeod::PlanetFixedPosition, [26](#)
- alt\_lat\_long\_state.cc, [33](#)
- alt\_lat\_long\_state.hh, [33](#)
- altitude
  - jeod::AltLatLongState, [17](#)
- altlatlong\_type
  - jeod::NorthEastDown, [22](#)
- AltLatLongState
  - jeod::AltLatLongState, [16](#)
- AltLatLongType
  - jeod::NorthEastDown, [19](#)
- build\_ned\_orientation
  - jeod::NorthEastDown, [20](#)
- cart\_coords
  - jeod::PlanetFixedPosition, [29](#)
- cart\_to\_ellip
  - jeod::PlanetFixedPosition, [26](#)
- cart\_to\_spher
  - jeod::PlanetFixedPosition, [27](#)
- class\_declarations.hh, [34](#)
- domain\_error
  - jeod::PlanetFixedMessages, [24](#)
- ellip\_coords
  - jeod::PlanetFixedPosition, [30](#)
- ellip\_to\_cart
  - jeod::PlanetFixedPosition, [27](#)
- elliptical
  - jeod::NorthEastDown, [19](#)
- get\_data
  - jeod::AltLatLongState, [16](#)
- get\_elliptic\_parameters
  - jeod::PlanetFixedPosition, [27](#)
- initialize
  - jeod::PlanetFixedPosition, [27](#)
- invalid\_request
  - jeod::PlanetFixedMessages, [24](#)
- jeod, [13](#)
- jeod::AltLatLongState, [15](#)
- ~AltLatLongState, [16](#)
- altitude, [17](#)
- AltLatLongState, [16](#)
- get\_data, [16](#)
- JEOD\_CLASS\_ESTABLISH\_FRIENDS2, [16](#)
- latitude, [17](#)
- longitude, [17](#)
- set\_data, [16](#)
- jeod::NorthEastDown, [18](#)
- ~NorthEastDown, [19](#)
- altlatlong\_type, [22](#)
- AltLatLongType, [19](#)
- build\_ned\_orientation, [20](#)
- elliptical, [19](#)
- JEOD\_CLASS\_ESTABLISH\_FRIENDS2, [20](#)
- ned\_frame, [22](#)
- NorthEastDown, [19](#), [20](#)
- operator=, [20](#)
- set\_ned\_trans\_states, [20](#)
- spherical, [19](#)
- undefined, [19](#)
- update\_from\_cart, [21](#)
- update\_from\_ellip, [21](#)
- update\_from\_spher, [21](#)
- jeod::PlanetFixedMessages, [23](#)
- domain\_error, [24](#)
- invalid\_request, [24](#)
- JEOD\_CLASS\_ESTABLISH\_FRIENDS2, [24](#)
- operator=, [24](#)
- PlanetFixedMessages, [23](#)
- jeod::PlanetFixedPosition, [25](#)
- ~PlanetFixedPosition, [26](#)
- cart\_coords, [29](#)
- cart\_to\_ellip, [26](#)
- cart\_to\_spher, [27](#)
- ellip\_coords, [30](#)
- ellip\_to\_cart, [27](#)
- get\_elliptic\_parameters, [27](#)
- initialize, [27](#)
- JEOD\_CLASS\_ESTABLISH\_FRIENDS2, [28](#)
- Max\_iteration\_limit, [30](#)
- planet, [30](#)
- PlanetFixedPosition, [26](#)
- Small\_radius\_limit, [30](#)
- spher\_to\_cart, [28](#)
- sphere\_coords, [31](#)
- update\_from\_cart, [28](#)
- update\_from\_ellip, [29](#)
- update\_from\_spher, [29](#)

- JEOD\_CLASS\_ESTABLISH\_FRIENDS2
  - jeod::AltLatLongState, [16](#)
  - jeod::NorthEastDown, [20](#)
  - jeod::PlanetFixedMessages, [24](#)
  - jeod::PlanetFixedPosition, [28](#)
- latitude
  - jeod::AltLatLongState, [17](#)
- longitude
  - jeod::AltLatLongState, [17](#)
- MAKE\_PLANETFIXED\_MESSAGE\_CODE
  - planet\_fixed\_messages.cc, [36](#)
- Max\_iteration\_limit
  - jeod::PlanetFixedPosition, [30](#)
- Models, [11](#)
- ned\_frame
  - jeod::NorthEastDown, [22](#)
- north\_east\_down.cc, [34](#)
- north\_east\_down.hh, [35](#)
- NorthEastDown, [11](#)
  - jeod::NorthEastDown, [19](#), [20](#)
- operator=
  - jeod::NorthEastDown, [20](#)
  - jeod::PlanetFixedMessages, [24](#)
- planet
  - jeod::PlanetFixedPosition, [30](#)
- planet\_fixed\_messages.cc, [35](#)
  - MAKE\_PLANETFIXED\_MESSAGE\_CODE, [36](#)
- planet\_fixed\_messages.hh, [36](#)
- planet\_fixed\_posn.cc, [36](#)
- planet\_fixed\_posn.hh, [37](#)
- PlanetFixed, [11](#)
- PlanetFixedMessages
  - jeod::PlanetFixedMessages, [23](#)
- PlanetFixedPosition
  - jeod::PlanetFixedPosition, [26](#)
- PlanetFixedPosn, [12](#)
- set\_data
  - jeod::AltLatLongState, [16](#)
- set\_ned\_trans\_states
  - jeod::NorthEastDown, [20](#)
- Small\_radius\_limit
  - jeod::PlanetFixedPosition, [30](#)
- spher\_to\_cart
  - jeod::PlanetFixedPosition, [28](#)
- sphere\_coords
  - jeod::PlanetFixedPosition, [31](#)
- spherical
  - jeod::NorthEastDown, [19](#)
- undefined
  - jeod::NorthEastDown, [19](#)
- update\_from\_cart
  - jeod::NorthEastDown, [21](#)
  - jeod::PlanetFixedPosition, [28](#)
- update\_from\_ellip
  - jeod::NorthEastDown, [21](#)
  - jeod::PlanetFixedPosition, [29](#)
- update\_from\_spher
  - jeod::NorthEastDown, [21](#)
  - jeod::PlanetFixedPosition, [29](#)
- Utils, [11](#)